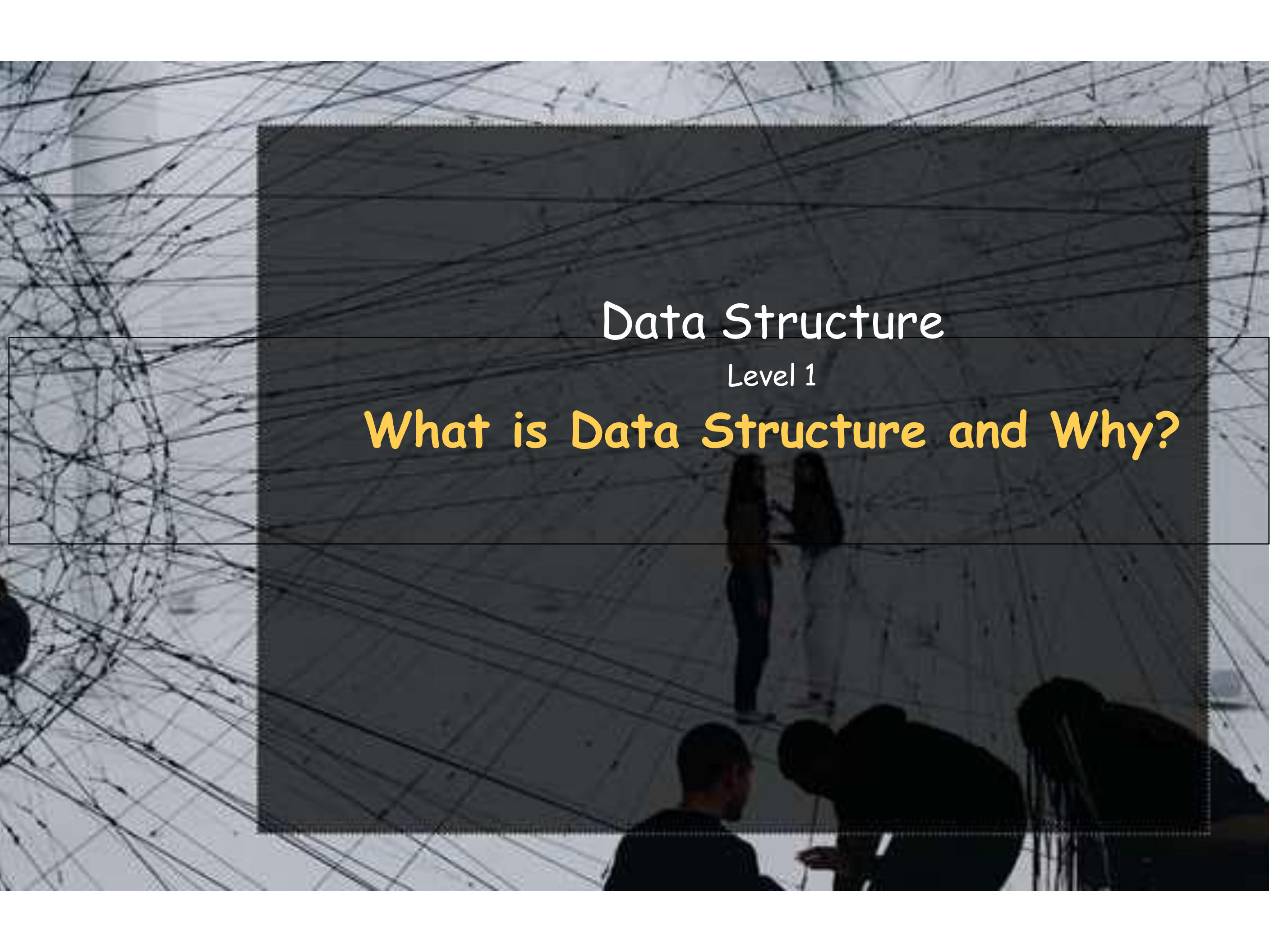


Data Structure

Level 1

What is Data Structure
and Why?

The background of the slide features a complex, abstract network of thin, dark lines crisscrossing a light gray background, resembling a web or a data structure. In the lower portion of the image, there are dark silhouettes of several people, some standing and some leaning over, suggesting a collaborative work environment or a meeting. A large, dark gray rectangular area is overlaid on the right side of the image, serving as a backdrop for the text.

Data Structure

Level 1

What is Data Structure and Why?

Definition: Program

- ◆ A **Program** is a complete solution written in a programming language to solve a specific problem.
- ◆ It is commonly expressed as:

Comparison

Program = Algorithms + Data Structures

Note:

An algorithm alone is not enough; without a proper data structure, the program may be inefficient or impractical.



Explanation: Algorithm

- ◆ An **Algorithm** is a step-by-step logical procedure used to solve a particular problem.
- ◆ It defines **what to do**, but not **how data is stored**.



Example:

Steps to find the largest number in a list.

Definition: Data Structure

- ◆ A **Data Structure** is a way of organizing, managing, and storing data items and their relationships inside a program so they can be used efficiently.

Interpretation:

Data structures decide **how data lives in memory** and **how easily it can be accessed or modified**.

Explanation: Why Data Structures Matter?

- ◆ Data structures affect:
 - Program efficiency
 - Memory usage
 - Speed of execution
 - Ease of data processing

Note:

Choosing the wrong data structure can make a correct algorithm slow or unusable.

Definition: Role of Data Structures

- ◆ A data structure is not only used for organizing data, but also for:
 - Processing data
 - Retrieving data
 - Storing data
 - Accessing data efficiently

Result:

Better performance and cleaner program design.

Explanation: Data Structures & Program Design

- ◆ Data structures influence both:
 - **Structural design** (how the program is built)
 - **Functional design** (how the program works)
- ⊘ **Warning:**
 - Ignoring data structure selection often leads to:
 - High time complexity
 - Wasted memory
 - Poor scalability

Definition: Type of Data Structures

- ◆ There are **basic** and **advanced** data structures used in almost every software system.

- **Basic Data Structures**

- Arrays
 - Linked Lists
 - Stacks
 - Queues

- **Advanced Data Structures**

- Trees
 - Graphs
 - Hash Tables
 - Heaps

 Note:

Each type has its own advantages, disadvantages, and use cases.



Question

- ◆ How do you identify the best data structure for a task?

■ Procedures

- 1 Analyze the problem
- 2 Identify operations needed (search, insert, delete)
- 3 Consider time complexity
- 4 Consider memory constraints
- 5 Choose the most efficient structure



Comparison

■ Algorithm

- ✓ Defines the step-by-step logic to solve a problem
- ✓ Focuses on *what to do*
- ✓ Independent of how data is stored
- ✓ Measured mainly by time complexity
- ✓ Cannot work alone without data

■ Data Structure

- ✓ Organizes and stores data efficiently
- ✓ Focuses on *how data is stored and accessed*
- ✓ Directly affects program performance
- ✓ Measured by time and space complexity
- ✓ Supports and optimizes algorithms

Summary

- ✓ A **Program** is built using algorithms and data structures
- ✓ Algorithms define the logic
- ✓ Data structures define how data is stored and managed
- ✓ Proper data structure selection improves efficiency
- ✓ Every real-world program depends heavily on data structures

Thank you 😊